

8. Develop a program using **malloc()**, **calloc()**, **realloc()**, and **free()**. Monitor memory allocation behavior and identify memory leaks using **Valgrind**

CODE:

```
advaith@DESKTOP-0E1OBMR: X + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    a = (int*)malloc(5 * sizeof(int));
    b = (int*)calloc(5, sizeof(int));
    a = (int*)realloc(a, 10 * sizeof(int));

    free(a);
    free(b);

    return 0;
}
```

```
advaith@DESKTOP-0E1OBMR:~$ gcc malloc.c -o program
advaith@DESKTOP-0E1OBMR:~$ ./program
Memory allocated using malloc: 0x5cc7f6dfe050
Memory allocated using calloc: 0x5cc7f6dfe030
Memory reallocated using realloc: 0x5cc7f6dfe050
Memory freed successfully.
advaith@DESKTOP-0E1OBMR:~$
```

OUTPUT: